

Development fit

6 seeds; observed next-answer loss
Choose estimator parameters

freeze

Held-out grid

6 personas x 2 simulator families
x 20 seeds = 240 histories / condition

Eleven conditions; 30 virtual days; eight concepts

Count / BKT / PFA x constant / conditional / value timing = nine
Two bounds: oracle timing and never intervene

Inside each learner history

Estimator + timing

Update estimate from observed
answers; choose an intervention
or no action.

action

Simulated learner

Hidden mastery, receptivity
and forgetting determine
subsequent responses.

observed response only

truth

Separate scorer

Compare estimates with hidden state: MSE, wasted-message fraction,
final hidden mastery. Only the oracle bound can use hidden state to act.